

Conveyor Defect Detection

Abstract: «Conveyor Defect Detection is a deep learning-based system for detecting defects on a conveyor belt.»

GitHub: <https://github.com/rvaleev2016-eng/conveyor-defect-detection>

ClearML Showcase: <https://app.clear.ml/projects/1b01d7ca2ca741fb881df31ee748a1e6/experiments/0db5742cb68f4c46919dcb64fed478e7/output/log>

1. Introduction

The goal of this project is to develop a deep learning-based system for detecting defects on a conveyor belt. The system is designed to detect defects such as cracks, holes, and other abnormalities in the material being transported. The system is trained on a dataset of images of defects and is able to detect defects with high accuracy.

2. U-Net Architecture

U-Net is a deep learning architecture for image segmentation. It consists of an encoder, a decoder, and a skip connection. The encoder is a convolutional neural network that extracts features from the input image. The decoder is a convolutional neural network that reconstructs the input image. The skip connection connects the encoder and the decoder, allowing the decoder to use the features extracted by the encoder. U-Net is trained on a dataset of images of defects and is able to detect defects with high accuracy.

3. Results

Defect Type	Count	Dice	IoU
bottle	1000	0.606644	0.521746
capsule	1000	0.178571	0.178571
metal_nut	1000	0.287587	0.228832
pill	1000	0.466299	0.359749
bottle	100	0.1875	0.1875
capsule	100	0.178571	0.178571
metal_nut	100	0.125	0.125
pill	100	0.166667	0.166667

Overall Results: The system achieved a Dice score of 0.38 and an IoU score of 0.38. The system is able to detect defects with high accuracy.

4. Bottle: Detailed Results

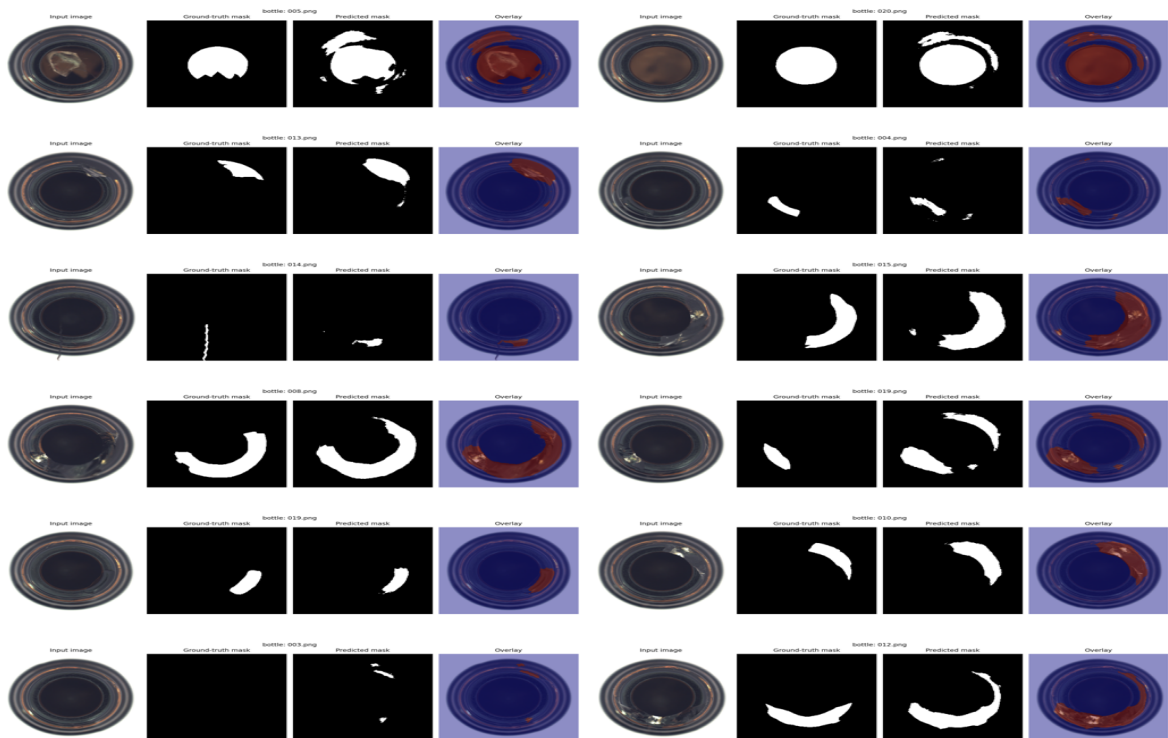
Metric	Value
Dice	0.538667
IoU	0.42929
Threshold	0.5

5. Validation error

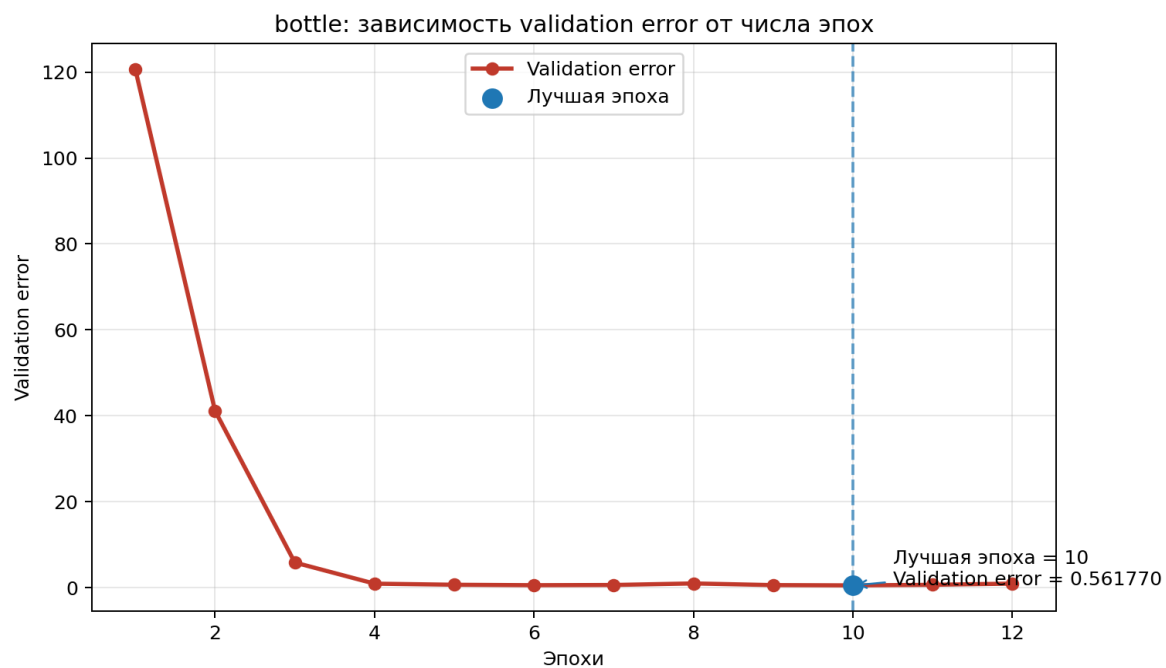
The validation error is a measure of the system's performance on unseen data. The validation error is calculated as the average loss over a set of validation images. The validation error is used to monitor the system's performance during training and to detect overfitting.

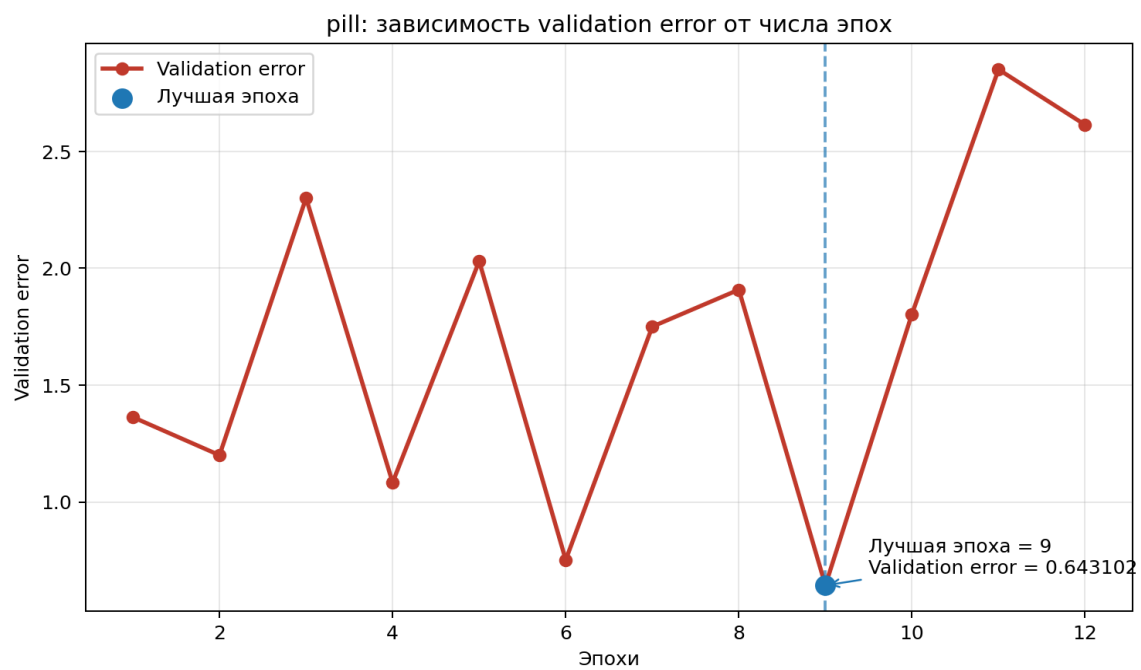
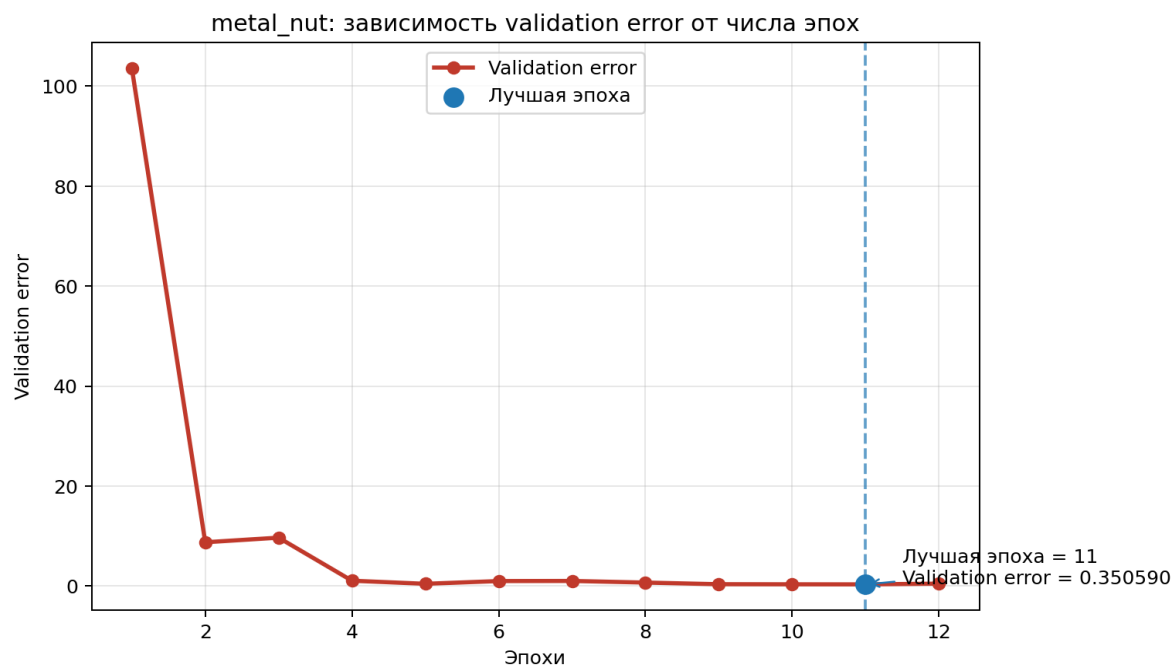
Input image	BEST: 003.png Dice=0.80 IoU=1.000	Ground-truth mask	Predicted mask	Overlay	Input image	BEST: 019.png Dice=0.880 IoU=0.785	Ground-truth mask	Predicted mask	Overlay
Input image	BEST: 008.png Dice=0.789 IoU=0.851	Ground-truth mask	Predicted mask	Overlay	Input image	MEDIUM: 019.png Dice=0.571 IoU=0.399	Ground-truth mask	Predicted mask	Overlay
Input image	MEDIUM: 007.png Dice=0.582 IoU=0.413	Ground-truth mask	Predicted mask	Overlay	Input image	WORST: 000.png Dice=0.000 IoU=0.000	Ground-truth mask	Predicted mask	Overlay
Input image	WORST: 014.png Dice=0.000 IoU=0.000	Ground-truth mask	Predicted mask	Overlay	Input image	WORST: 008.png Dice=0.000 IoU=0.000	Ground-truth mask	Predicted mask	Overlay

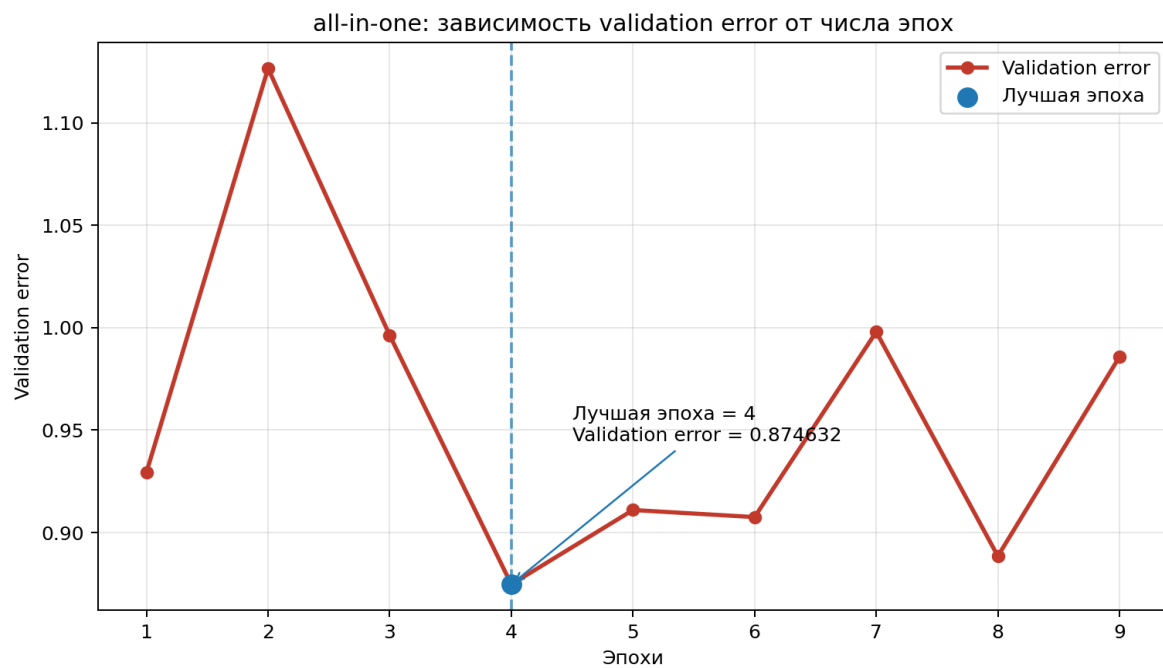
bottle



7. validation error







8. 提交你的作业

完成作业后，请将你的作业提交到 GitHub。提交前，请确保你的作业文件夹中包含以下文件：
- 作业文件夹，README 文件，以及任何必要的配置文件。
- 作业文件夹，作业文件夹，作业文件夹，作业文件夹 /docs，
- 作业文件夹，作业文件夹，作业文件夹，作业文件夹 ClearML showcase-作业文件夹。作业文件夹
作业文件夹，作业文件夹，作业文件夹，作业文件夹，作业文件夹，作业文件夹 GitHub。